CS 1300 - Homework 5

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GitHub repository: https://github.com/mimifantastic/hw5

Section 1: Multiple Choice

1. b) B
2. b) "odd"
3. b) The conditions are in the wrong order - amount > 100 should come before amount > 50
4. b) value_if_true if condition else value_if_false
5. a) True
6. a) Valid size
7. b) "blue" is a non-empty string (truthy), so the or expression is always True
8. b) It removes leading and trailing whitespace
9. c) An off-by-one error
10. b) 2-3 levels

Section 2: True or False

1. T
2. F
3. T
4. F
5. F

Section 3: Python Programs

Problem 1: Movie Ticket Pricing

problem1_movie_ticket.py

```
# Calculate the ticket price using age and matinee status.
def main():
    try:
        age = int(input("Enter your age: "))
    except ValueError:
        print("Error: age must be an integer.")
        return
    if age < 0:
        print("Error: age must be non-negative.")
        return

    answer = input("Is this a matinee showing? (yes/no): ")
    answer = answer.strip().lower()
    if answer not in ["yes", "no"]:
        print("Error: enter yes or no.")
        return
    matinee = True if answer == "yes" else False

    # The outer conditions select age; inner conditions select price.
    if age < 13:
        group = "Child"
        if matinee:
            price = 6.00
        else:
            price = 8.00
    elif age < 18:
        group = "Teen"
        if matinee:
            price = 7.00
        else:
            price = 10.00
    elif age < 65:
        group = "Adult"
        if matinee:
            price = 8.00
        else:
            price = 13.00
    else:
        group = "Senior"
        if matinee:
            price = 6.00
        else:
            price = 7.00

    print(f"Age group: {group}")
    print(f"Ticket price: ${price:.2f}")


if __name__ == "__main__":
    main()
```

Section 3: Python Programs

Problem 2: Input Validator

problem2_input_validator.py

```
# Collect every field before checking the profile.
student_id = input("Enter student ID: ").strip()
name = input("Enter full name: ").strip()
age_text = input("Enter age: ").strip()
major = input("Enter major: ").strip().upper()
errors = []

# Check each ID rule independently; slicing is safe for an empty ID.
if len(student_id) != 8:
    errors.append(
        f"Student ID must be exactly 8 characters (got {len(student_id)})"
    )
if not student_id[:1].isalpha():
    errors.append("Student ID must start with a letter")
if not student_id[1:].isdigit():
    errors.append("Student ID characters after the first must be digits")

if not name:
    errors.append("Name cannot be empty")
elif len(name) < 2:
    errors.append("Name must be at least 2 characters")

# Only check the range if conversion to an integer succeeds.
try:
    age = int(age_text)
    if not 16 <= age <= 99:
        errors.append("Age must be between 16 and 99")
except ValueError:
    errors.append("Age must be a valid integer")

if major not in ["CS", "IT", "CE", "DS"]:
    errors.append(f"Major must be one of: CS, IT, CE, DS (got {major})")

# Report all errors together, or display the accepted profile.
if errors:
    print("Profile has errors:")
    for error in errors:
        print(f"  - {error}")
else:
    print("Profile created successfully!")
    print(f"  Student ID: {student_id}")
    print(f"  Name:       {name}")
    print(f"  Age:        {age}")
    print(f"  Major:      {major}")
```

Section 3: Python Programs

Problem 3: Campus Cafe Menu

problem3_campus_cafe.py

```
# Use integer cents so all prices and tax round consistently.
def main():
    print("=" * 30)
    print("    CAMPUS CAFE ORDER SYSTEM")
    print("=" * 30)
    print("  1. Coffee   - $3.50")
    print("  2. Sandwich - $6.00")
    print("  3. Salad    - $5.50")
    print("  4. Combo    - $8.00")
    print("  5. Exit")
    print("=" * 30)
    choice = input("Enter your choice (1-5): ").strip()
    if choice == "5":
        print("Goodbye!")
        return
    if choice not in ["1", "2", "3", "4"]:
        print("Invalid choice. Please select 1-5.")
        return

    # Set the base price, then customize inside the selected branch.
    if choice in ["1", "4"]:
        if choice == "1":
            item, cents = "Coffee", 350
        else:
            item, cents = "Combo (Sandwich + Coffee)", 800
        size = input("Coffee size (small/medium/large): ").strip().lower()
        if size == "medium":
            cents += 100
        elif size == "large":
            cents += 200
        elif size != "small":
            print("Invalid size. Defaulting to Small.")
            size = "small"
        item += f" ({size.title()})"
    elif choice == "2":
        item, cents = "Sandwich", 600
    else:
        item, cents = "Salad", 550
        dressing = input("Dressing (ranch/italian/vinaigrette/none): ")
        dressing = dressing.strip().lower()
        if dressing not in ["ranch", "italian", "vinaigrette", "none"]:
            print("Invalid dressing. Defaulting to none.")
            dressing = "none"
        item += f" (dressing: {dressing})"
```

Section 3: Python Programs

Problem 3: Campus Cafe Menu (continued)

problem3_campus_cafe.py

```
    # Both the sandwich and combo offer cheese.
    if choice in ["2", "4"]:
        cheese = input("Add cheese? (yes/no): ").strip().lower()
        while cheese not in ["yes", "no"]:
            print("Please enter yes or no.")
            cheese = input("Add cheese? (yes/no): ").strip().lower()
        if cheese == "yes":
            cents += 75
            item += " + Cheese"

    name = input("Enter your name: ").strip()
    while not name:
        print("Name cannot be empty.")
        name = input("Enter your name: ").strip()

    # Retry until quantity is an integer greater than zero.
    while True:
        try:
            quantity = int(input("How many? "))
            if quantity > 0:
                break
            print("Quantity must be positive.")
        except ValueError:
            print("Quantity must be a valid integer.")

    subtotal = cents * quantity
    # Round 7% tax to the nearest cent, rounding half a cent up.
    tax = (subtotal * 7 + 50) // 100
    total = subtotal + tax
    print("=" * 30)
    print("        ORDER RECEIPT")
    print("=" * 30)
    print(f"  Customer: {name}")
    print(f"  Item:     {item}")
    print(f"  Quantity: {quantity}")
    print(f"  Unit Price: ${cents / 100:.2f}")
    print(f"  Subtotal:   ${subtotal / 100:.2f}")
    print(f"  Tax (7%):   ${tax / 100:.2f}")
    print(f"  Total:      ${total / 100:.2f}")
    print("=" * 30)
    print("Thank you for your order!")


if __name__ == "__main__":
    main()
```